

Amente Bekele

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Work Experience

Google

3.5 years

Senior Software Engineer, Site Reliability Engineering

January 2023–Present

- Design, implement, and optimize systems to ensure the availability, durability, and scalability of Google's planet-scale distributed data storage infrastructure.
- Lead on-call incident response and root-cause analysis (RCA) for critical storage outages, defining and enforcing strict Service Level Objectives (SLOs) and Error Budgets.
- Architect automation pipelines using Go and Python to eliminate operational toil, reducing manual intervention in storage node provisioning and recovery.
- Drive capacity planning and performance tuning across highly distributed clusters, leveraging telemetry and data analysis to optimize resource utilization and realize millions in infrastructure costs savings.

Okta

1.5 years

Senior Software Engineer

June 2021–December 2022

- Engineered and operated a scalable, Kubernetes-based multi-cloud platform that served as the backbone for mission-critical CI/CD and testing workloads.
- Hardened platform infrastructure and optimized resource allocation, ensuring high availability for engineering teams executing 100,000+ daily automated tests and deployments.
- Implemented comprehensive observability and alerting for the multi-cloud platform to proactively detect and mitigate cluster degradation.

IBM

6 years

Advisory Software Engineer

May 2015–June 2021

- Led the development of the IBM Cognos Analytics mobile app (React Native) and its associated backend microservices (TypeScript/Node.js, Java).
- Drove the end-to-end design and production deployment of IBM Watson Analytics for Social Media, transforming a conceptual design into a highly scalable cloud solution.
- Engineered the backend social data analytics pipeline using Apache Spark, Kafka, and Zookeeper, processing 10+ TB of streaming data daily with high fault-tolerance.
- Championed production reliability by architecting automated CI/CD pipelines for Kubernetes using Jenkins and Docker, reducing deployment times by days across multiple teams.
- Collaborated directly with stakeholders to architect resilient technical solutions, prioritizing system observability and continuous optimization for cloud workloads.

IBM

16 months

Extreme Blue Technical Intern and Software Developer Co-op

May 2013–Aug 2014

- Co-developed a Social Media Analytics prototype on IBM BlueMix (Node.js, Java, Cloudant), collaborating directly with a client to define requirements and deliver a successful proof of concept.
- Pitched the business value proposition of the analytics prototype to senior executives, including the CEO of IBM.
- Streamlined developer workflows by building a Python-based Eclipse plugin for internal build automation and an advanced log analysis tool (Eclipse RCP) for the L3 support team.

Carleton University

3 years

Research Assistant, Teaching Assistant, Lab Assistant

July 2012–December 2015

- Developed a web application using PHP and MySQL to query a protein-protein interaction prediction database for the Bioinformatics Lab.
- Modernized the curriculum for a senior-level microcontroller course by porting laboratory materials and exercises from Intel 8051 to ARM Cortex M4.
- Mentored 20 students as a Teaching Assistant for a highly project-driven course on ARM-Cortex M microcontrollers (SYSC 4906).

Grype Solutions

6 months

Software Developer

July 2012–January 2013

- Developed the shopping cart UI for a web-based Point of Sale (POS) terminal software using C# and ASP.NET.
- Implemented robust hardware integration interfaces to support receipt printers and barcode scanners.

Education

Academic Qualifications.....

- **Carleton University** **Ottawa**
○ *MASc Electrical and Computer Engineering, Data Science Specialization,* *2016–September 2018*
Thesis: Developed a Neonatal Respiratory Rate Monitoring system using a pressure-sensitive mat, applying data analysis, digital signal processing, and machine learning for NICU patient monitoring.
- **Carleton University** **Ottawa**
○ *B.Eng. Computer Systems Engineering , Graduated May 2016* *2011–2016*

Technical skills

- **Programming Languages:** Java, JavaScript/TypeScript, Python, C, C++, C#, Scala, Go
- **Cloud Platforms & Infrastructure:** AWS, IBM Cloud, GCP, Digital Ocean, Kubernetes, Docker
- **Data Analytics & ML:** Kafka, Zookeeper, Spark, MATLAB, SciPy, TensorFlow
- **Hardware:** ARM Cortex M, Zigbee Networks, Bluetooth, Wifi, Hardware Protocols and Interfaces: I2C, SPI, USB

Notable projects as a consultant

- **SnekTek SolarDashboard:** Engineered and maintained C++ firmware and a Node.js web application portal for a solar power monitoring product, scaling it to support hundreds of active monthly subscribers.
- **Gluecontrols.com:** Developed C-based firmware for an advanced distance-type pattern controller system used in packaging applications across North and South America.
- **WAVEFLO Leak Detection System:** Re-architected a legacy C++ valve leak detection system into a modern C# application for tablet and USB platforms, including the successful re-implementation of a proprietary DSP algorithm for the power generation industry.
- **MeVPMDD:** Shipped a globally utilized cross-platform mobile application and backend (Apache Cordova/AngularJS) for tracking menstrual cycles and PMDD symptoms.

Interests and Extracurricular school activity

- **IEEE Carleton University Student Branch (Chair, Computer Society Chair, Webmaster):** Served in multiple leadership roles. Organized technical competitions, developed an award-winning branch website (1st place globally, 2012), and secured the IEEE Computer Society Richard Merwin Award (\$1K) and IEEE Canadian Foundation scholarship (\$10K).
- **Carleton University Robotics Club (Workshop Coordinator, Project Lead):** Led a 10-person engineering team to design a satellite system, ranking 8th globally in the 2013 CanSat Competition. Mentored subsequent teams (ranked 2nd and 3rd) and coordinated robotics workshops for 12 local high schools.
- **AlertBuddy (Engineering Capstone Project):** Designed an emergency alarm audio detection algorithm using DSP and Artificial Neural Networks for a wearable device. Awarded the Dr. Walter Chudobiak Entrepreneurship Award (\$10K).
- **Amateur Radio, Advanced License VA3AXB**

Research

- Authored 10+ peer-reviewed publications with over 130 citations focused on digital signal processing and machine learning for patient monitoring applications. A link to my Google Scholar profile can be found below:

